Homework Signal 5

Week 7

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1 DFT & DTFT

Problem 1. Discrete Fourier Transform (DFT):

1.1 Given the DFT spectrum $X[k]$, express the corresponding time-domain signal $x[n]$ in terms of its constituent real sinusoids.

TO SUBMIT

1.1.1 $X[0] = 3, X[1] = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}, X[2] = -2, X[3] = \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\left(\frac{2\pi}{N}\right)kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned} X[k] &= \left\{3, \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}, -2, \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right\} \\ &= \left\{3, \cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right), -2, \cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right)\right\} \\ X[k] &= \left\{3, e^{-j\frac{\pi}{4}}, -2, e^{j\frac{\pi}{4}}\right\} \end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned} x[n] &= \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\left(\frac{2\pi}{N}\right)kn} \\ &= \frac{1}{N}\left(X[0]e^{j\left(\frac{2\pi}{N}\right)k(0)} + X[1]e^{j\left(\frac{2\pi}{N}\right)k(1)} + X[2]e^{j\left(\frac{2\pi}{N}\right)k(2)} + X[3]e^{j\left(\frac{2\pi}{N}\right)k(3)}\right) \\ &= \frac{1}{4}\left(3e^{j\left(\frac{2\pi}{4}\right)(0)n} + e^{-j\frac{\pi}{4}}e^{j\left(\frac{2\pi}{4}\right)(1)n} - 2e^{j\left(\frac{2\pi}{4}\right)(2)n} + e^{j\frac{\pi}{4}}e^{j\left(\frac{2\pi}{4}\right)(3)n}\right) \\ &= \frac{1}{4}\left(3e^{0} + e^{j\left(\frac{\pi}{4}\right)(2n-1)} - 2e^{j\pi n} + e^{j\left(\frac{\pi}{4}\right)(6n+1) - j(2\pi)}\right) \\ &= \frac{1}{4}\left(3 - 2(-1)^n\right) + \frac{1}{4}\left(e^{j\left(\frac{\pi}{4}\right)(2n-1)} + e^{j\left(\frac{\pi}{4}\right)(6n+1-8n)}\right) \\ &= \frac{1}{4}\left(3 - 2(-1)^n\right) + \frac{1}{4}\left(e^{j\left(\frac{\pi}{4}\right)(2n-1)} + e^{-j\left(\frac{\pi}{4}\right)(2n-1)}\right) \\ &= \frac{1}{4}\left(3 - 2(-1)^n\right) + \frac{1}{2}\cos\left(\frac{\pi}{4}(2n-1)\right) \\ x[n] &= \frac{1}{4}\left(3 - 2(-1)^n\right) + \frac{1}{2}\cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right) \end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$\boxed{x[n] = \frac{1}{4}\left(3 - 2(-1)^n\right) + \frac{1}{2}\cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right)}$$

TO SUBMIT

1.1.2 $X[0] = -2, X[1] = \sqrt{3} + j, X[2] = 3, X[3] = \sqrt{3} - j$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned}
X[k] &= \left\{-2, \sqrt{3} + j, 3, \sqrt{3} - j\right\} \\
&= \left\{-2, 2\left(\frac{\sqrt{3}}{2} + j\frac{1}{2}\right), 3, 2\left(\frac{\sqrt{3}}{2} - j\frac{1}{2}\right)\right\} \\
&= \left\{-2, 2\left(\cos\left(\frac{\pi}{6}\right) + j\sin\left(\frac{\pi}{6}\right)\right), 3, 2\left(\cos\left(-\frac{\pi}{6}\right) + j\sin\left(-\frac{\pi}{6}\right)\right)\right\} \\
X[k] &= \left\{-2, 2e^{j\frac{\pi}{6}}, 3, 2e^{-j\frac{\pi}{6}}\right\}
\end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned}
x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn} \\
&= \frac{1}{N}\left(X[0]e^{j\left(\frac{2\pi}{N}\right)k(0)} + X[1]e^{j\left(\frac{2\pi}{N}\right)k(1)} + X[2]e^{j\left(\frac{2\pi}{N}\right)k(2)} + X[3]e^{j\left(\frac{2\pi}{N}\right)k(3)}\right) \\
&= \frac{1}{4}\left(-2e^{j\left(\frac{2\pi}{4}\right)(0)n} + 2e^{j\frac{\pi}{6}}e^{j\left(\frac{2\pi}{4}\right)(1)n} + 3e^{j\left(\frac{2\pi}{4}\right)(2)n} + 2e^{-j\frac{\pi}{6}}e^{j\left(\frac{2\pi}{4}\right)(3)n}\right) \\
&= \frac{1}{4}\left(-2e^{0} + 2e^{j\left(\frac{\pi}{6}\right)(3n+1)} + 3e^{j\pi n} + 2e^{j\left(\frac{\pi}{6}\right)(9n-1)-j(2\pi)}\right) \\
&= \frac{1}{4}\left(-2 + 3(-1)^n\right) + \frac{1}{4}\left(2e^{j\left(\frac{\pi}{6}\right)(3n+1)} + 2e^{j\left(\frac{\pi}{6}\right)(9n-1-12n)}\right) \\
&= \frac{1}{4}\left(-2 + 3(-1)^n\right) + \frac{1}{2}\left(e^{j\left(\frac{\pi}{6}\right)(3n+1)} + e^{-j\left(\frac{\pi}{6}\right)(3n+1)}\right) \\
&= \frac{1}{4}\left(-2 + 3(-1)^n\right) + \cos\left(\frac{\pi}{6}(3n+1)\right) \\
x[n] &= \frac{1}{4}\left(-2 + 3(-1)^n\right) + \cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right)
\end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$\boxed{x[n] = \frac{1}{4}\left(-2 + 3(-1)^n\right) + \cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right)}$$

TO SUBMIT

1.1.3 $X[0] = 1, X[1] = 2 - j2\sqrt{3}, X[2] = -3, X[3] = 2 + j2\sqrt{3}$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned}
X[k] &= \left\{1, 2 - j2\sqrt{3}, -3, 2 + j2\sqrt{3}\right\} \\
&= \left\{1, 4\left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right), -3, 4\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right\} \\
&= \left\{1, 4\left(\cos\left(-\frac{\pi}{3}\right) + j\sin\left(-\frac{\pi}{3}\right)\right), -3, 4\left(\cos\left(\frac{\pi}{3}\right) + j\sin\left(\frac{\pi}{3}\right)\right)\right\} \\
X[k] &= \left\{1, 4e^{-j\frac{\pi}{3}}, -3, 4e^{j\frac{\pi}{3}}\right\}
\end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned}
x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn} \\
&= \frac{1}{N}\left(X[0]e^{j\left(\frac{2\pi}{N}\right)k(0)} + X[1]e^{j\left(\frac{2\pi}{N}\right)k(1)} + X[2]e^{j\left(\frac{2\pi}{N}\right)k(2)} + X[3]e^{j\left(\frac{2\pi}{N}\right)k(3)}\right) \\
&= \frac{1}{4}\left((1)e^{j\left(\frac{2\pi}{4}\right)(0)n} + 4e^{-j\frac{\pi}{3}}e^{j\left(\frac{2\pi}{4}\right)(1)n} + (-3)e^{j\left(\frac{2\pi}{4}\right)(2)n} + 4e^{j\frac{\pi}{3}}e^{j\left(\frac{2\pi}{4}\right)(3)n}\right) \\
&= \frac{1}{4}\left(e^0 + 4e^{j\left(\frac{\pi}{6}\right)(3n+2)} - 3e^{j\pi n} + 4e^{j\left(\frac{\pi}{6}\right)(9n-2)-j(2\pi)}\right) \\
&= \frac{1}{4}(1 - 3(-1)^n) + \frac{1}{4}\left(4e^{j\left(\frac{\pi}{6}\right)(3n+2)} + 4e^{j\left(\frac{\pi}{6}\right)(9n-2-12n)}\right) \\
&= \frac{1}{4}(1 - 3(-1)^n) + \left(e^{j\left(\frac{\pi}{6}\right)(3n+2)} + e^{-j\left(\frac{\pi}{6}\right)(3n+2)}\right) \\
&= \frac{1}{4}(1 - 3(-1)^n) + 2\cos\left(\frac{\pi}{6}(3n+2)\right) \\
x[n] &= \frac{1}{4}(1 - 3(-1)^n) + 2\cos\left(\frac{\pi}{2}n + \frac{\pi}{3}\right)
\end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$\boxed{x[n] = \frac{1}{4}(1 - 3(-1)^n) + 2\cos\left(\frac{\pi}{2}n + \frac{\pi}{3}\right)}$$

1.2 Show that if $x[n]$ is real sequence, $X[N-k] = X^*[k]$

Solution. Using the DFT definition:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\left(\frac{2\pi}{N}\right)kn}$$

To find $X[N-k]$:

$$\begin{aligned}
X[N-k] &= \sum_{n=0}^{N-1} x[n]e^{-j\left(\frac{2\pi}{N}\right)(N-k)n} \\
&= \sum_{n=0}^{N-1} x[n]e^{-j\left(2\pi n - \frac{2\pi}{N}kn\right)} \\
&= \sum_{n=0}^{N-1} x[n]e^{-j2\pi n}e^{j\left(\frac{2\pi}{N}\right)kn} \\
&= \sum_{n=0}^{N-1} x[n]e^{j\left(\frac{2\pi}{N}\right)kn} \quad \text{(since } e^{-j2\pi n} = 1 \text{ for integer } n\text{)} \\
&= \sum_{n=0}^{N-1} x[n]\left(e^{-j\left(\frac{2\pi}{N}\right)kn}\right)^* \\
&= \left(\sum_{n=0}^{N-1} x[n]e^{-j\left(\frac{2\pi}{N}\right)kn}\right)^* \quad \text{(since } x[n] \text{ is real)} \\
X[N-k] &= X^*[k]
\end{aligned}$$

Thus, we have shown that if $x[n]$ is a real sequence, then $X[N-k] = X^*[k]$.

$$\boxed{x[n] \text{ is real sequence} \implies X[N-k] = X^*[k]}$$

Problem 2. Discrete-Time Fourier Transform (DTFT):

2.1 Use the properties of the discrete-time transform to determine $X(e^{j\omega})$ for the following sequences

TO SUBMIT

2.1.1 $x[n] = \left(\frac{1}{3}\right)^{|n|}$

Solution. Using the DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Substituting the given sequence:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^{|n|} e^{-j\omega n} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^{n} e^{-j\omega n} + \sum_{n=-\infty}^{-1} \left(\frac{1}{3}\right)^{|n|} e^{-j\omega n} \end{aligned}$$

By substituting $m = -n$ in the second sum, we got:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^{n} + \sum_{m=1}^{\infty} \left(\frac{1}{3}\right)^{m} e^{j\omega m} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^{n} + \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m} e^{j\omega m} - \left(\frac{1}{3}\right)^{0} e^{j\omega(0)} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^{n} + \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^{m} e^{j\omega m} - 1 \end{aligned}$$

Then, using the geometric series formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{1-\frac{1}{3}e^{-j\omega}} + \frac{1}{1-\frac{1}{3}e^{j\omega}} - 1 \\ &= \frac{1-\frac{1}{3}e^{j\omega} + 1 - \frac{1}{3}e^{-j\omega}}{\left(1-\frac{1}{3}e^{-j\omega}\right)\left(1-\frac{1}{3}e^{j\omega}\right)} - 1 \\ &= \frac{\left[2 - \cancel{\frac{1}{3}\left(e^{j\omega}+e^{-j\omega}\right)}\right] - \left[1 - \cancel{\frac{1}{3}\left(e^{j\omega}+e^{-j\omega}\right)} + \frac{1}{9}\right]}{1-\frac{1}{3}\left(e^{j\omega}+e^{-j\omega}\right)+\frac{1}{9}} \\ &= \frac{2-\frac{10}{9}}{1-\frac{2}{3}\cos\omega+\frac{1}{9}} \\ &= \frac{\frac{8}{9}}{\frac{10}{9}-\frac{2}{3}\cos\omega} \\ X(e^{j\omega}) &= \frac{4}{5-3\cos\omega} \end{aligned}$$

Thus, the DTFT of the sequence $x[n] = \left(\frac{1}{3}\right)^{|n|}$ is:

$$\boxed{X(e^{j\omega}) = \frac{4}{5-3\cos\omega}}$$

TO SUBMIT

2.1.2 $x[n] = a^n \cos(\Omega_0 n) \cdot u[n]$, $|a| < 1$

Solution. Using the DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Substituting the given sequence:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} a^n \cos(\Omega_0 n) \cdot u[n]e^{-j\omega n} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} a^n \cos(\Omega_0 n)e^{-j\omega n} \end{aligned}$$

Using the Euler's formula $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, we got:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{\infty} a^n \left(\frac{e^{j\Omega_0 n} + e^{-j\Omega_0 n}}{2}\right) e^{-j\omega n} \\ &= \frac{1}{2}\sum_{n=0}^{\infty} a^n e^{j\Omega_0 n} e^{-j\omega n} + \frac{1}{2}\sum_{n=0}^{\infty} a^n e^{-j\Omega_0 n} e^{-j\omega n} \\ X(e^{j\omega}) &= \frac{1}{2}\sum_{n=0}^{\infty} \left(ae^{j(\Omega_0-\omega)}\right)^n + \frac{1}{2}\sum_{n=0}^{\infty} \left(ae^{-j(\Omega_0+\omega)}\right)^n \end{aligned}$$

Then, using the geometric series formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{1}{1-ae^{j(\Omega_0-\omega)}} + \frac{1}{2} \cdot \frac{1}{1-ae^{-j(\Omega_0+\omega)}} \\ &= \frac{1}{2} \cdot \frac{1-ae^{-j(\Omega_0+\omega)} + 1 - ae^{j(\Omega_0-\omega)}}{\left(1-ae^{j(\Omega_0-\omega)}\right)\left(1-ae^{-j(\Omega_0+\omega)}\right)} \\ X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{2 - a\left(e^{j(\Omega_0-\omega)} + e^{-j(\Omega_0+\omega)}\right)}{1 - a\left(e^{j(\Omega_0-\omega)} + e^{-j(\Omega_0+\omega)}\right) + a^2e^{-2j\omega}} \end{aligned}$$

Consider the value of $e^{j(\Omega_0-\omega)} + e^{-j(\Omega_0+\omega)}$:

$$\begin{aligned} e^{j(\Omega_0-\omega)} + e^{-j(\Omega_0+\omega)} &= e^{j\Omega_0}e^{-j\omega} + e^{-j\Omega_0}e^{-j\omega} \\ &= e^{-j\omega}\left(e^{j\Omega_0} + e^{-j\Omega_0}\right) \\ e^{j(\Omega_0-\omega)} + e^{-j(\Omega_0+\omega)} &= 2e^{-j\omega}\cos\Omega_0 \end{aligned}$$

Substituting back:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{2 - 2ae^{-j\omega}\cos\Omega_0}{1 - 2ae^{-j\omega}\cos\Omega_0 + a^2e^{-2j\omega}} \\ X(e^{j\omega}) &= \frac{1 - ae^{-j\omega}\cos\Omega_0}{1 - 2ae^{-j\omega}\cos\Omega_0 + a^2e^{-2j\omega}} \end{aligned}$$

Thus, the DTFT of the sequence $x[n] = a^n \cos(\Omega_0 n) \cdot u[n]$ is:

$$\boxed{X(e^{j\omega}) = \frac{1}{2}\left(\frac{1}{1-ae^{j(\Omega_0-\omega)}} + \frac{1}{1-ae^{-j(\Omega_0+\omega)}}\right) = \frac{1 - ae^{-j\omega}\cos\Omega_0}{1 - 2ae^{-j\omega}\cos\Omega_0 + a^2e^{-2j\omega}}}$$

TO SUBMIT

2.1.3 $x[n] = (n+1)a^n \cdot u[n]$, $|a| < 1$

Solution. Using the Linearity property of DTFT:

$$X(e^{j\omega}) = \text{DTFT}\left\{(n+1)a^n \cdot u[n]\right\} = \text{DTFT}\left\{na^n \cdot u[n]\right\} + \text{DTFT}\left\{a^n \cdot u[n]\right\}$$

Define $y[n] = a^n \cdot u[n]$, and $z[n] = ny[n]$, we get:

$$X(e^{j\omega}) = \text{DTFT}\left\{z[n]\right\} + \text{DTFT}\left\{y[n]\right\}$$

From the property of DTFT for $y[n] = a^n \cdot u[n]$ $|a| < 1$:

$$Y(e^{j\omega}) = \text{DTFT}\left\{y[n]\right\} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

Using the Differentiation property of DTFT:

$$\text{DTFT}\left\{nx[n]\right\} = j\frac{dX(e^{j\omega})}{d\omega}$$

where $X(e^{j\omega}) = \text{DTFT}\left\{x[n]\right\}$.
Thus, we have:

$$\begin{aligned}
Z(e^{j\omega}) &= j\frac{dY(e^{j\omega})}{d\omega} \\
&= j\frac{d}{d\omega}\left(\frac{1}{1 - ae^{-j\omega}}\right) \\
&= j \cdot \left(\frac{0 \cdot (1 - ae^{-j\omega}) - 1 \cdot \left(-a(-j)e^{-j\omega}\right)}{(1 - ae^{-j\omega})^2}\right) \\
&= j \cdot \left(\frac{-jae^{-j\omega}}{(1 - ae^{-j\omega})^2}\right) \\
Z(e^{j\omega}) &= \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2}
\end{aligned}$$

Combining both results:

$$\begin{aligned}
X(e^{j\omega}) &= Z(e^{j\omega}) + Y(e^{j\omega}) \\
&= \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2} + \frac{1}{1 - ae^{-j\omega}} \\
&= \frac{ae^{-j\omega} + (1 - ae^{-j\omega})}{(1 - ae^{-j\omega})^2} \\
X(e^{j\omega}) &= \frac{1}{(1 - ae^{-j\omega})^2}
\end{aligned}$$

Thus, the DTFT of the sequence $x[n] = (n+1)a^n \cdot u[n]$ is:

$$\boxed{X(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})^2}}$$